

National Academy of Science and Technology
(Affiliated to Pokhara University)
Dhangadhi, Kailali

Pre-University Examination

Level: Bachelor
Program: B.E. Civil/Computer
Course: Principles of Engineering Economic Analysis
Semester: VII- Fall

Year : 2023
F.M. : 100
P.M. : 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions

1.a) "Knowledge of engineering economics helps in decision making process". Justify it by the principles of engineering economics. [8]

b) Explain manufacturing cost and non-manufacturing cost with example. [7]

2.a) You make 10 annual deposits of 1,000 each into a bank account paying 5% interest per year. The first deposit will be made one year from today.

How much money can be withdrawn from this account after 10 years?

What is the present worth of that sum? [7]

b) A piece of new equipment has been proposed by engineers to increase the productivity of a certain manual welding operation. The investment cost of equipment is Rs 1, 00,000 and has market value of Rs. 20,000 at the end of five years. The equipment will generate Rs. 32,000 per year after operation costs. The company is willing to accept the project if it will earn at least 20% per year. Evaluate IRR of this equipment and suggest whether the equipment is feasible or not? [8]

3.a) Calculate PW of the following 2 projects using MARR 12% and suggest whether the project is feasible or not? [7]

	A	B
Investment	4,00,000	6,00,000
Annual Revenue	30,000	35,000
Annual O and M	3,000	4,000
Life	8	8
SV	4000	7000

b) From the following information, conduct scenario analysis based on FW formulation. Assume $I = 2, 23,000$; $MARR = 13.5\%$; and life of project is 5 years. Also give your remarks based on result of different scenarios. [8]

Variable considered	Worst case scenario	Most likely scenario	Best case scenario
Annual sales	86000	110000	137000
Annual variable cost	37000	40000	38000
Annual fixed cost	21000	20000	18000
Salvage value	40000	50000	60000

4. a) A mobile company is taking quotations for purchase, installation and operation of microwave towers for long period. If MARR is 15%, determine the best alternative project using repeatability assumption and PW method. [7]

Quotations	Project A	Project B
Equipment cost (Rs)	6500000	5800000
Installation (Rs)	1500000	2000000
Annual maintenance (Rs)	100000	125000
Annual extra charge (Rs)	0	50000
Salvage value (Rs)	0	0
Life (yrs)	15	10

- b) An asset has installed value of 45000. $SV = 0$. It is classed as a 5-year property. Determine approximate MACRS depreciation schedule. Over 6 years it is estimated to generate revenue of Rs. 23000 per year with annual operating cost 7300. Required rate of return = 15% after tax. Tax rate = 40%. Evaluate after tax IRR with annual worth method. [8]

- 5.a) Assuming that a firm has a tax rate 40%, compute after tax cost of the following:

- i) A 10 years bond of face value Rs. 1,000 sold at 5% premium with a 10% coupon rate.
- ii) A common stock of Rs. 100 par value, selling at Rs. 110 and paying a Rs. 20 dividend expected to grow at the rate of 5%. [7]
- b) What is financial ratio? Explain major financial ratios that can be applied in decision making process in business. [8]

- 6.a) What Do you mean by Public Private Partnership? Explain in brief. [7]
- b) Define ecological footprint. Explain the concept of sustainable development. [8]

7. Write short notes on: (Any two)

[2*5=10]

- a. Mutually exclusive, contingent and independent project
- b. FIRR and EIRR
- c. Balance sheet

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Pre-University Examination

Level: Bachelor

Semester: VII - Fall

Year : 2023

Program: BE Computer

F.M. : 100

Course: Artificial Intelligence

P.M. : 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group – A

Attempt all questions.

1. ✓ What is the Turing Test in AI? Criticize the performance of the Turing Test to measure the intelligence of the machine. 7
2. Define quantifier. Explain different inference rules for quantifiers. 8
3. ✓ Why NLP is difficult. Explain the steps in the process of NLP. 7
4. Define Alpha-Beta Pruning? Explain an alpha cutoff and beta cutoff in detail. 7
5. How can you convert to conjunctive normal form? Explain all the steps with examples. 8
6. ✓ Why is a learning framework required? Explain about explanation-based machine learning. 8
7. ✓ Explain the hill-climb method with different limitations using suitable examples. 7
8. ✓ Model the tictac toe as a production system. 7
9. Explain the nonparametric learning methods. Describe the k-means learning methods with examples. 8
10. ✓ What are the steps for developing an expert system? 7
11. ✓ What are the steps to solve the constraint satisfaction problem? Solve the following crypt arithmetic problem. 8
DONALD
+ GERALD

ROBERT
12. Represent the following sentence in first-order logic: 8
✓ i. Every student smiles.

- ii. Everyone walks or talks.
- iii. Every student who loves Ravi is happy.
- iv. If anyone cheats, everyone suffers.

13. Write short notes on: (Any Two)

2*5

- i. ✓ Overview of Neural Network
- ii. Fuzzy Learning
- iii. ✓ Breadth First Search

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Pre-University Examination

Level: Bachelor

Semester: VII - Fall

Programme: BE Computer

Course: **Image Processing and Pattern Recognition**

Year : 2023

F.M. : 100

P.M. : 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Digital Image Processing? Explain the Fundamental steps of Digital Image Processing. 7

- b) List out the applications of Histogram. Perform linear stretching on the given image. 8

Gray level	0	1	2	3	4	5	6	7
Frequency	0	0	50	60	50	20	10	0

2. a) How can you perform Zooming on image? Explain with example. 7

- b) What is Hadamard transform? Explain DFT in detail. 8

“OR”

Explain Haar Transform with algorithm and its properties.

3. a) Explain, Opening and Closing with example? 8

- b) How the Image is Restored after degradation. Explain. 7

4. a) Explain Lossless and Lossy compression technique in detail? 8

- b) List out the Image segmentation techniques based on similarity and explain anyone. 7

5. a) Construct Huffman tree for each gray level given and find the efficiency. 7

Gray Level	0	1	2	3	4	5	6	7
No. of pixels	400	1350	659	2034	816	2560	250	1500

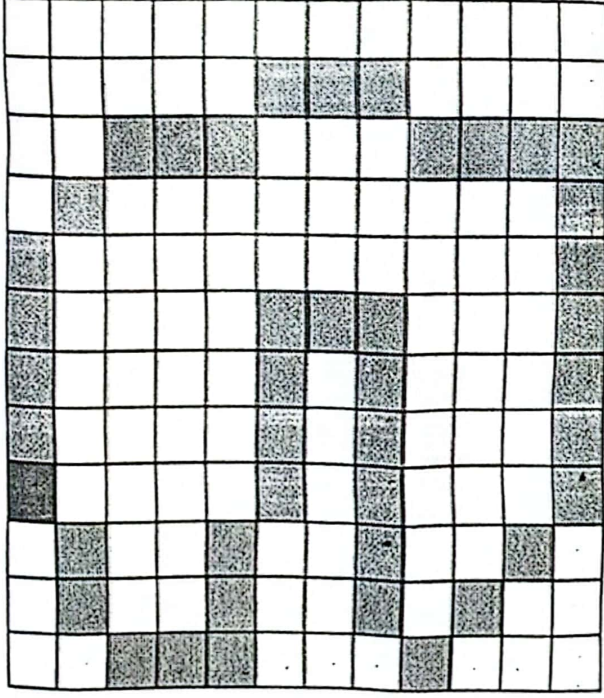
- b) Explain line and Spot detection? 8

“OR”

List and explain any one type of Edge Detection technique.

6. a) Find the shape number of the following Image?

7



b) Explain Artificial Neural Network. List and explain any one application of NN in Image processing.

8

7. Write short notes on following (Any Two)

5x2

- a) Sampling and Quantization
- b) Application of Pattern Recognition
- c) Fourier Transform

Good Luck!!!

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Pre University Examination

Level: Bachelor Semester – Fall_VII Year : 2023
Programme: BE Computer Full Marks: 100
Course: Computer Network Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Discuss the merits and demerits of computer network. Compare star, bus, ring and mesh topology. [7]
b) Explain the working principle of different types of network devices: hub, switch, bridge, and router. [8]
2. a) Explain circuit switching mechanism. Explain the terms throughput, latency, bandwidth and jitter. [7]
b) What is error control. Explain CRC method of error detection with a suitable example. [8]
3. a) What are intermediate devices in a network? Specify some of the physical, datalink and network layer devices. Describe how fiber optics cable provides faster, reliable and efficient means for data transmission. [7]
b) What are the components of IPV4 frame format? Differentiate between IPV4 and IPV6. [4+4]
4. a) NAST College is planning to design 8 different department for BE Computer, BCA, BBA, BE Civil, Exam, Research, Account and library. Each department consists of 30 computers with IP address 192.168.218.0. Explain the process how will you allocate the IP address and subnet for each of the departments using above IP. [8]

- b) Discuss about Interior and Exterior Routing protocols. What is count to infinity problem in distance vector. [7]
5. a) What is congestion in a network. Explain about Leaky bucket algorithm with suitable diagram and algorithm. [8]
b. List out and explain TCP client/server socket flow with suitable diagram. [7]
6. a) What is data encryption? Discuss the role of symmetric and asymmetric key cryptography with a suitable example.. [2+6]
b. What do you mean by network security? Explain RSA algorithm. [7]
7. Write short notes on: (Any two) [2x5=10]
✓ a) Frame Relay
b) Firewalls
c) OSI and TCP/IP

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Pre-University Examination

Level: Bachelor

Semester : VII_Fall

Year: 2023

Program: B.E. Computer

F.M.: 100

Course: ICT Project Management

P. M.: 45

Time : 3 Hrs.

Candidates are required to give their answer in their own words as for as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is a project system? Describe the project environment. [8]
 b) Define organizational structure. Which organizational structure is suitable for projects? Why? [7]
2. a) Define OPA. Describe various types of organizational process assets. [7]
 b) Describe in detail, what are the necessary contents in a project charter. [8]
3. a) What is WBS? Illustrate a WBS for a project of library software design. [8]
 b) A project consists of the following activities. [5]

Activity	Predecessors	Time estimates		
		Optimistic	Most likely	pessimistic
A	-	3	6	9
B	-	2	3	4
C	A	2	4	6
D	A	3	4	5
E	A	1	3	5
F	C	2	3	4
G	D	3	4	11
H	B, D, E	4	5	6
I	H	2	2	2
J	I, G, F	2	3	4

- a. Draw a project network diagram.

- b. Identify critical path and critical activities.
- c. Calculate the project duration.

4. a) Explain the following schedule development tools and concepts: Bar charts, Milestone chart and Gantt chart. [10]
b) Define budget. Explain about the various techniques of determining budget.

5. a) Define cost of quality. How can project managers control quality? Explain with basic quality control tools. [7]
b) Define human resource management. Describe different processes of human resource management. [8]

6. a) What is communication? Describe the basic communication model of project. [7]
b) What is risk in the project? Describe about various types of risks in project management. [8]

7. Write short notes on: [Any Two] [2x5]
a) Stakeholders of projects
b) Public Procurement
c) Quality Assurance

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Pre-University Examination

Level: Bachelor

Semester: VII_Fall

Year : 2023

Program: BE Computer

Full Marks: 100

Course: Mobile Application Development Technology
(Elective II)

Pass Marks : 45

Time : 3hrs

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

- 1 a) List out different mobile OSes. Explain the popular mobile operating system with architectural diagram. 8
b) Explain the Architecture of Android VM and Runtime (Dalvik and ART). 7
- 2 a) Define Sensor. Explain different types of sensors in brief. 7
b) Define the fundamental building blocks of android application. Also Explain Activity life-cycle diagram in details. 8
- 3 a) Define Fragment? Explain the use of fragment with its life-cycle diagram. 8
b) Explain the importance of AndroidManifest.xml file. Write down manifest file for default activity and set some permissions on it. 7
- 4 a) List different types of menus in android. Explain the importance of UI and Navigation in android. Write down the code to generate simple popup menu. 8
b) Explain the importance of Threads, AsyncTask in android. Illustrate with code block. 7
- 5 a) Define notification. Explain different form of notification with examples. 8
b) Write down the steps to signing and publishing apps to play store. 7

- 6 a) Write down SQLite database code to create, read, update and delete records of table tbl_students (id, name, email, mobile, status) in separate methods. 10
- b) List out the 3rd party UI /UX libraries. Illustrate when to use them and why? 5

7. Write down short notes (on any two)

- a) Camera API
- b) Multi-touch and Gestures
- c) The Content provider classes

2 x 5 = 10